

ASTERION

Mathematics and scientific research with checkable evidence

A local research platform for exploring mathematical questions, testing scientific models and preserving reproducible evidence. This public repository explains selected results and how to review them.

Selected recorded results

Public exact-arithmetic checker. The separate MIT Bernstein checker includes 40 recorded polynomials. A fresh download passed 13 test methods, 400 synthetic conversion cases, 2,000 exact point comparisons and 240 rejected semantic mutations.

Partial formal proof foundation. A retained Lean run accepted two generic coefficient-bound theorems, 40 polynomial identities and 40 denominator bounds. The full putrit theorem and Python checker are not formally verified.

Retained quantum measurements. 6,144 saved IBM Quantum and Quantum Inspire shots produced 72 outward confidence enclosures. Coverage is conditional on stationary IID shots within each circuit/marginal; physical drift was not established.

Evidence limits

A local reliability reference pilot checked 28,672 simulated paths and 210 exact reference intervals. It uses a standard concentration corollary; the condition for a larger campaign was not met. No novelty, hardware superiority or quantum advantage is established.

Useful next contribution

Independent review of one narrow claim, a concrete prior-art gap or a useful contribution to an existing benchmark.

MIRANDA handles defensive cybersecurity and forensic research. ASTERION handles mathematics and scientific experiments.

<https://github.com/QuantumParadox/ASTERION>

Overview only; full platform not relicensed. The separate Bernstein checker has its own MIT license.